SIMATS ENGINEERING

ASSESSMENT TOOL 3 – CO2 AT3

Debugging & Optimisation Task

Student Name	Bhavatharani H
Register Number	1921260256
Course Code	CSA0709
Course Name	Computer Networks
Course Outcome	CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using a simulation tool, for efficient network design and topology (BL3)
Tools	Cisco Packet Tracer, Wireshark
Marks	5 questions × 6 marks = 30 marks

Question 1: IPv4 Workstation Configuration (6 marks)

Given: IP 192.168.10.65, Mask 255.255.255.192 (/26), Gateway 192.168.10.1. The workstation cannot reach hosts in 192.168.10.0/26.

1. Subnet of the configured IP

A /26 mask has 6 host bits, so the block size is 256 – 192 = 64. The subnets in 192.168.10.0 are .0, .64, .128 and .192. The address .65 falls in the block starting at .64, so it belongs to **192.168.10.64/26**, not to 192.168.10.0/26.

Item	192.168.10.0/26 (target subnet)	192.168.10.64/26 (workstation's subnet)
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable hosts	62	62

2. Validity of the IP address and gateway

- **IP 192.168.10.65:** a valid host address (the first usable address of 192.168.10.64/26), but it is in the *wrong subnet* for the intended LAN.
- **Gateway 192.168.10.1: invalid** for this host. It lies in 192.168.10.0/26, while the host is in 192.168.10.64/26. A default gateway must be in the same subnet as the host's interface, otherwise ARP for the gateway never succeeds.

3. Configuration errors identified

- Subnet mismatch: host IP (.64/26) and gateway/peers (.0/26) are in different subnets.
- Because the subnets differ, the workstation treats 192.168.10.x (x < 64) as remote and sends the traffic to a gateway it cannot reach at Layer 2, so communication fails.

4. Optimised configuration

Option A (preferred – place the host in the intended subnet 192.168.10.0/26):

Parameter	Corrected value
IP address	192.168.10.10 (any unused address in .2 – .62)
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1 (first usable address, router interface)

Option B (keep the host in 192.168.10.64/26): IP 192.168.10.65, mask 255.255.255.192, and the gateway changed to a router interface in the same subnet, e.g. **192.168.10.126**. The IP .65 is then moved to a different address such as .66 if the router uses .65.

5. Total usable host addresses after correction

Usable hosts = $2^n - 2 = 2^6 - 2 = 64 - 2 =$ **62 usable hosts** per /26 subnet.

Question 2: IPv6 Addressing Debug (6 marks)

Given: Device A: 2001:db8::ff00:42:8329/64 Device B: 2001:db8:0:0:1::1/64. Both are on the same network but cannot communicate.

1. Full (expanded) notation

Device	Compressed	Expanded
A	2001:db8::ff00:42:8329	2001:0db8:0000:0000:0000:ff00:0042:8329
B	2001:db8:0:0:1::1	2001:0db8:0000:0000:0001:0000:0000:0001

In A, the `::` replaces 3 zero groups (5 groups are written, 8 needed). In B, `::` replaces 2 zero groups (6 groups written).

2. Network prefix of each device (/64 = first 4 groups)

Device	Network prefix (first 64 bits)	Interface ID (last 64 bits)
A	2001:0db8:0000:0000 → 2001:db8::/64	0000:ff00:0042:8329
B	2001:0db8:0000:0000 → 2001:db8::/64	0001:0000:0000:0001

3. Same subnet?

Yes. Both addresses have the identical /64 prefix **2001:db8:0:0::/64**. B's extra '1' sits in the fifth group, which is part of the *interface ID* (bits 65–80), not the subnet ID. It is a common mistake to read it as a subnet ID, which would wrongly suggest a prefix mismatch.

4. Addressing / prefix mismatch

- There is **no prefix mismatch**: both are /64 on 2001:db8::/64, so on-link reachability does not depend on the prefix.
- The addresses are valid but irregular (non-sequential interface IDs, hard to manage and easy to mistype/misread as in the 5th group). Since the prefixes match, the real fault must be checked at other layers: wrong VLAN/port, interface or link down, IPv6 not enabled on one device, ICMPv6/Neighbor Discovery blocked by a firewall/ACL, or a duplicate address.
- Verify with: `show ipv6 interface brief`, `ping 2001:db8:...`, `show ipv6 neighbors`.

5. Optimised configuration

Device	Optimised address	Prefix
Device A	2001:db8:0:1::10	/64 (2001:db8:0:1::/64)
Device B	2001:db8:0:1::20	/64 (2001:db8:0:1::/64)
Router (gateway)	2001:db8:0:1::1	/64

```
ipv6 unicast-routing
interface g0/0
 ipv6 address 2001:db8:0:1::1/64
 no shutdown
! On hosts: static or SLAAC with prefix 2001:db8:0:1::/64
```

6. Number of available host addresses

A /64 leaves 128 – 64 = 64 host bits, so the subnet holds 2^{64} = **18,446,744,073,709,551,616** (about 1.8×10^{19}) addresses. IPv6 has no broadcast address, so effectively all of them are assignable (only the all-zeros Subnet-Router anycast address is reserved).

Question 3: Subnet Allocation Analysis and VLSM Optimisation (6 marks)

Note on assumptions: the question does not list the host requirements per department. The following realistic requirements are assumed for the analysis: **Engineering 50, Sales 25, HR 12, Accounts 10** hosts (97 hosts in total). The method applies to any other figures.

1. Existing allocation of 192.168.1.0/24 (using /26, /27, /28)

Dept	Subnet	Network	Broadcast	Host range	Usable	Needed	Unused / shortfall
Sales	/26	192.168.1.0	192.168.1.63	.1 – .62	62	25	37 unused
Engineering	/27	192.168.1.64	192.168.1.95	.65 – .94	30	50	**20 short**
HR	/28	192.168.1.96	192.168.1.111	.97 – .110	14	12	2 unused
Accounts	/26	192.168.1.128	192.168.1.191	.129 – .190	62	10	52 unused

Totals: usable = 62 + 30 + 14 + 62 = 168; needed = 97; **unused = 91 (excluding the Engineering shortfall of 20)**. The block 192.168.1.112 – .127 (16 addresses) is left stranded between HR and Accounts.

2. Inefficient allocations and mistakes found

- **Engineering (/27, 30 hosts)** is undersized: needs 50 hosts, so it needs at least a /26.
- **Sales (/26)** and **Accounts (/26)** are oversized: a /26 gives 62 hosts for 25 and 10 hosts, wasting 37 and 52 addresses.
- **HR (/28)** is nearly full (only 2 spare), leaving no room for growth.
- Fixed-size thinking (plain subnetting) instead of VLSM: subnets were not sized by requirement, and the largest requirement was not allocated first.
- A 16-address gap (.112 – .127) is fragmented and unused. Note that a /27 cannot start at .112 because /27 blocks must start at multiples of 32 (.0, .32, .64, .96, .128 ...), so the gap cannot be reused for a /27.

3. Optimised VLSM scheme (allocate largest first)

Dept	Hosts needed	Mask	Network	Host range	Broadcast	Usable	Spare
Engineering	50	/26 (255.255.255.192)	192.168.1.0	.1 – .62	192.168.1.63	62	12
Sales	25	/27 (255.255.255.224)	192.168.1.64	.65 – .94	192.168.1.95	30	5
HR	12	/28 (255.255.255.240)	192.168.1.96	.97 – .110	192.168.1.111	14	2
Accounts	10	/28 (255.255.255.240)	192.168.1.112	.113 – .126	192.168.1.127	14	4
Free pool	–	/25	192.168.1.128	.129 – .254	192.168.1.255	126	future use / WAN /30 links

Result: all four departments are now satisfied (Engineering's 20-host shortfall is fixed). Allocated usable addresses fall from 168 to 120 (needed 97, unused 23), and the whole upper half, **192.168.1.128/25** (128 addresses), is free for growth, new VLANs or /30 router-to-router links. Subnets are contiguous, aligned on their block boundaries and non-overlapping, which also keeps route summarisation possible (192.168.1.0/25 summarises all four).

Question 4: IPv6 Addressing Debug (6 marks)

Given: Device A: 2001:db8::ff00:42:8329/64 Device B: 2001:db8:0:0:1::1/64. Both are on the same network but cannot communicate.

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Remark: Question 4 uses the same IPv6 addresses as Question 2, so the analysis and corrected addressing are identical. A Packet Tracer test is to ping between the corrected addresses and capture the ICMPv6 Neighbor Solicitation/Advertisement exchange in Wireshark to confirm on-link reachability.

Question 5: Routing Table Debug and Optimisation (6 marks)

Given routing table:

Destination network	Next hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default route	10.0.0.1

Packets destined for **192.168.2.100** are not reaching the intended network.

1. Matching entry (longest-prefix match)

Route	Does 192.168.2.100 match?	Prefix length
192.168.1.0/24 (192.168.1.0 – .255)	No (third octet is 1)	–
192.168.2.0/24 (192.168.2.0 – .255)	**Yes**	/24
Default 0.0.0.0/0	Yes (matches everything)	/0

Two routes match; the router chooses the longest prefix, so **192.168.2.0/24 (/24)** wins over the default route (/0).

2. Next-hop router selected

10.0.0.3.

3. Does the destination belong to the intended subnet?

Yes. 192.168.2.100 is within 192.168.2.0/24 (network 192.168.2.0, broadcast 192.168.2.255, hosts .1 – .254).

4. Routing configuration errors

- The router's route *selection* is correct, so the destination is not mis-matched. The problem is in the entry's **next hop**: 10.0.0.3 is either not the router that actually connects to 192.168.2.0/24 (the next hops for the two networks may be swapped or mistyped), or it is unreachable (down, no ARP entry, wrong interface).
- The next-hop router may also lack a route back to the source network (return path missing), so replies are lost.
- The default route is written as 'Default Route' without an explicit 0.0.0.0/0 form and no exit interface is given, so it depends on recursive lookup of 10.0.0.1.
- Check with `show ip route`, `ping 10.0.0.3`, `traceroute 192.168.2.100`, and confirm the physical/Layer-2 link toward 10.0.0.3.

5. Optimised routing table

Destination	Mask	Next hop / exit	Remark
192.168.1.0/24	255.255.255.0	10.0.0.2 (verified reachable)	Specific static route
192.168.2.0/24	255.255.255.0	Correct router for that LAN (confirmed), via exit interface	Fix/verify next hop; add return route on that router
0.0.0.0/0	0.0.0.0	10.0.0.1	Explicit default route for all unknown destinations

```
ip route 192.168.1.0 255.255.255.0 10.0.0.2
ip route 192.168.2.0 255.255.255.0 <verified next hop>
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Further improvements: use specific /24 routes (not broad ones) so longest-prefix matching works predictably; summarise only when both networks share the same next hop (here they use different next hops, so summarising 192.168.0.0/22 would be wrong); remove duplicate or unused entries; add a floating static or a routing protocol (OSPF) for redundancy.

6. Route used when no destination matches

The **default route (0.0.0.0/0) via 10.0.0.1**, for example for 192.168.3.50 or 8.8.8.8.